

MEMORANDUM

TO: Senior Operations Manager
FROM: Amos Njagi Ndungo, Software & Systems Engineer
DATE: July 17, 2026
SUBJECT: Automated Operational Efficiency Tracking Solution (Pilot Phase)

1. CONTEXT: THE NEED FOR AUTOMATED EFFICIENCY TRACKING

In modern retail and supply operations, manual performance auditing is a costly bottleneck. Relying on managers to manually compile daily sheets, calculate yields, and flag underperforming units delays crucial interventions. By the time a spreadsheet highlights a struggling location, days or weeks of sub-optimal performance have already dented our bottom line. Automating this process ensures real-time oversight, allowing leadership to shift focus from tedious data computation to immediate, proactive problem-solving.

2. SYSTEM OVERVIEW: WHAT WE BUILT IN PART A

We engineered a lightweight, automated tracking engine using Python—a versatile programming language designed for rapid automation. Rather than requiring manual calculations, the engine processes operational data and instantly evaluates performance using two main components:

- **Efficiency Calculation:** A self-contained calculation block (referred to in technical teams as a *"function"*) that automatically divides actual production or sales output by set targets, producing an objective percentage score.
- **Status Classification:** An automated rule-set that instantly grades each unit's health:
 - **NORMAL** : Efficiency of 90% or above (operating at peak standard).
 - **WARNING** : Efficiency between 70% and 89.9% (minor bottlenecks detected).
 - **CRITICAL** : Efficiency below 70% (requires immediate manager intervention).

3. BUSINESS INSIGHTS: REAL-WORLD DEMONSTRATION

To prove the concept, we ran the automated tool using live branch data from Kirinyaga County (Mwea, Kianyaga, and Sagana branches). The system instantly surfaced the following results without human error:

- **Mwea Branch:** **CRITICAL** (14.0% efficiency) — Severe underperformance requiring immediate operational audit.
- **Kianyaga Branch:** **CRITICAL** (47.3% efficiency) — Significantly below threshold; likely suffering from major supply or staffing disruptions.
- **Sagana Branch:** **WARNING** (80.7% efficiency) — Stable but experiencing minor bottlenecks that restrict optimal output.

The Value Contrast: Under a manual tracking model, compiling this data would take hours of searching and manual calculation. Our automated system completed these assessments and flagged the high-risk branches in less than a second, establishing a standard baseline for performance tracking.

4. ACTION PLAN: DEPARTMENTAL DEPLOYMENT

To drive immediate value in your department, we recommend integrating this tool directly into your weekly review workflows. Instead of manually reviewing static reports, we can schedule the system to pull sales and logistics data every Monday morning. The engine will instantly generate an aggregated dashboard, highlighting the exact locations operating in **CRITICAL** or **WARNING** statuses. This allows your management team to allocate resources, deploy support, and resolve bottlenecks before they impact weekly targets.